DATA SCIENCE LABORATORY MANUAL

Module II ? Python for Data Science

Topic: Pandas

OVERVIEW & LEARNING OBJECTIVES

Series and DataFrame data structures, indexing, filtering, column manipulations, and merging/joining tables.

KEY TOPIC CONCEPTS & METHODOLOGY

- Understanding Pandas in modern Data Science workflows.
- Detailed breakdown: Series and DataFrame data structures, indexing, filtering, column manipulations, and merging/joining tab
- Mathematical formulas, code examples, and laboratory execution guidelines.
- Common pitfalls, edge cases, and best practices in practical applications.

LABORATORY PRACTICE / IMPLEMENTATION NOTES

1. Follow standard Python and Data Science coding practices.
2. Document all exploratory findings and statistical hypotheses.
3. Keep datasets and outputs organized in your repository workspace.